April 1, 2025

Hon. Michelle L. Phillips
Secretary
New York Public Service Commission
3 Empire State Plaza
Albany, NY 12223-1350

RE: Case 22-E-0236: Proceeding to Establish Alternatives to Traditional Demand-Based Rate Structures for Commercial Electric Vehicle Charging

Comments of the Vehicle-Grid Integration Council (VGIC)
on the Demand Charge Rebate and Commercial Managed Charging Program

Introduction

The Vehicle-Grid Integration Council (VGIC)[1] is a 501(c)(6) membership-based advocacy group committed to advancing the role of electric vehicles (EVs) and vehicle-grid integration (VGI) through policy development, education, outreach, and research. VGIC supports the transition to decarbonized transportation and electric sectors by ensuring the value from flexible EV charging and discharging is recognized and compensated in support of a more reliable, affordable, and efficient electric grid. VGIC appreciates the opportunity to provide comments on the Demand Charge Rebate and Commercial Managed Charging Program (CMCP). Additionally, VGIC also reiterates a proposal regarding bidirectional charging that it has provided across several relevant EV proceedings in New York: compensation for vehicle-to-grid (V2G) exports.

Commercial Managed Charging Program

[1] VGIC member companies and supporters include American Honda Motor Co., Inc., Fermata Energy, Ford Motor Company, General Motors, Nissan Group of North America, Bidirectional Energy, BorgWarner, ChargeScape, dcbel, Eaton Corporation, Emporia Corp., EnergyHub, EV.Energy, Gravity, GridBeyond, GridWiz, Hyundai Cradle, Innovation Core SEI, Kaluza, Landis + Gyr, LG Innotek, Mercedes-Benz, Orange Charger, Peak Power, Stellantis N.V., Sunrun, Tesla, The Mobility House, Toyota Motor North America, Inc., WeaveGrid, Clean Power Alliance, Holy Cross Energy, Los Angeles Department of Water & Power, and Sacramento Municipal Utility District. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (https://www.vgicouncil.org/)

VGIC strongly supports the extension of the downstate utilities' Commercial Managed Charging Program. Particularly, VGIC is encouraged that the program is designed to incorporate customers both with and without separate meters, thereby facilitating participation from the broadest possible set of customers. This strategy is critical to unlocking a significant amount of regular peak load reduction from EV charging sites, a tool that will become increasingly valuable as New York's grid demands increased load flexibility.

Given the success of the downstate utilities' CMCP, VGIC reiterates its previous support within this proceeding for CMCP development for the remaining New York utilities. Managed charging offerings are increasingly commonplace among utilities across the country, as they offer an important tool not only to manage bulk system net peak load, but also more localized distribution system operational constraints.

Evolving Commercial Managed Charging Programs to Support Bidirectional Charging Solutions

In addition to extending and expanding existing CMCP efforts, VGIC strongly recommends the Commission direct the Joint Utilities to develop related incentives to support not only load reduction from commercial EVSE sites, but also vehicle battery discharge via bidirectional charging systems. These systems are increasingly available to commercial customers. For example, every major electric school bus manufacturer includes bidirectional charging capability. Meanwhile, bidirectional charging equipment, formerly offered by only Fermata Energy and BorgWarner, are now available from Tellus Power Green, Heliox, and InCharge, while Ford, General Motors, Wallbox, Tesla, and dcbel offer additional solutions currently tailored to light-duty vehicles and electric pickup trucks.

Bidirectional charging, especially in grid-parallel "vehicle-to-building" or "vehicle-to-grid" configurations, can significantly reduce the total operating costs of commercial charging sites, including school bus, refuse truck, and other fleet depot sites. In addition to reducing operating costs, these sites can serve as non-wires alternatives to mitigate distribution infrastructure needs and serve as an essential tool to help meet New York's energy needs in the coming years. Particularly, New York's aggressive climate and environmental goals will necessitate the retirement of polluting fossil fuel resources. Since utility-scale renewable energy development faces significant financial headwinds and transmission interconnection challenges, the state must leverage all available tools to ensure sufficient energy supply and reliability, especially in downstate areas. For example, NYISO recently had to delay the retirement of peaker plants in New York City to ensure reliability, noting that New York City would otherwise face an expected 446 MW of capacity deficiency beginning in mid-2025. Given the high real estate costs in downstate New York, VGI has immense potential to serve the state's needs since

EVs and the associated charging infrastructure will already be deployed as a result of the state's EV adoption goals. Unlike most other energy resources, unlocking bidirectional capabilities from these resources will not require the acquisition of additional real estate (i.e., compared to unidirectional charging). The NYISO Gold Book 2023 EV stock forecast includes 4,800 electric buses by 2025 and 28,800 electric buses by 2030. Modestly assuming that half are school buses with 120 kWh batteries and 40 kW chargers, school buses alone would total 96 MW of capacity and 288 MWh of energy in 2025, and 576 MW of capacity and 1.728 GWh of energy in 2030.

However, current infrastructure costs and export compensation mechanisms do not make investments in V2G-capable vehicles and chargers economically viable in New York. Additional policy intervention and market support, addressing both infrastructure costs and export compensation for V2G, are necessary to spur deployment and ensure that the emerging V2G resource can be leveraged to support New York's decarbonization goals in the coming years. VGIC appreciates the Commission's recent decision in Docket 18-E-0138 to require the Joint Utilities to update their VDER tariffs to include V2G, which will establish uniformity across the state regarding the eligibility of V2G resources to participate in VDER. Based on VGIC's estimates, the compensation under VDER is currently insufficient to warrant investments in V2G projects, especially in areas outside of ConEd territory. Using highly generous assumptions for a 120-kWh electric bus (i.e., batteries discharge all 120 kWh of stored energy across three hours of a CSRP call window for 20 days per month), fleets can earn a maximum of approximately \$8,400 per vehicle over three summer months in ConEd territory by participating in VDER. However, in National Grid territory, the corresponding best-case-scenario compensation drops to only approximately \$1,800 per vehicle per summer. While VGIC has not developed similar estimates for the remaining utilities, we understand the other IOUs' Value Stack credit rates are similar to or lower than those offered by National Grid, and therefore provide a similar or even lower compensation level for V2G projects. Given the relatively high upfront costs of developing and deploying bidirectional charging infrastructure and the enabling equipment, which is particularly high as the market is in a nascent stage of development, the (somewhat unrealistic) best-case-scenario VDER revenue in most New York utility territories (with the exception of ConEd) would be wholly insufficient to unlock this latent V2G capacity. Additionally, the CSRP and DLRP programs require participants to forego the DRV and LSRV components of VDER but only include a small number of events each summer, resulting in even lower compensation than what fleets can earn from the DRV component alone.

In contrast, a managed bidirectional charging program could be offered that would include much more attractive compensation mechanism for V2G customers. For example, National Grid's Massachusetts ConnectedSolutions Daily Dispatch program calls between 30 and 60 events per summer, each lasting two to three hours, and provides \$200 per kW per summer in compensation based on average performance. For comparison, again assuming 120 kWh of battery capacity, the compensation fleets can earn from V2G is between $8,000 and $12,000 per vehicle per summer in Massachusetts. Similarly, Rhode Island Energy's ConnectedSolutions program provides $300 per kW per summer, allowing a much smaller vehicle battery (Nissan's 60-kWh LEAF) on a much smaller charger (Fermata Energy's 15-kW FE-15) to earn, on average, approximately $4,000 per summer.

Based on these current program offerings, VGIC expects V2G investments on the east coast to likely concentrate toward New England states with ConnectedSolutions options. Even though revenue under the VDER tariff were comparable to the real-world results from V2G projects in New England, New York V2G site deployment will still likely lag, given the overall higher costs faced by fleets and technology providers in the state, especially in the downstate area. Without improvements to its V2G export compensation mechanism(s), New York risks being left out of the development of V2G projects and, instead, may become home to an immense amount of a new type of stranded asset: latent distributed energy storage capacity locked away behind unidirectional chargers.

VGIC proposes that the Commission establish a new managed bidirectional charging program in addition to the CMCP offering. This program could be similar to the Daily Dispatch program referenced above, to incentivize grid exports from V2G. The program could either 1) be an option in which fleets participate instead of receiving the DRV and LSRV credits under VDER, 2) provide additional compensation on top of all VDER components, or 3) provide an entirely non-rate alternative that is closely aligned with the CMCP approach and incentive. This strategy would mirror the Commission's reasoning for establishing the Term-DLM and Auto-DLM programs: spur investments into a specific customer resource that will benefit the grid. In the case of Term-DLM and Auto-DLM, the goal was to support more long-term, capital-intensive stationary storage projects and spur the emerging stationary energy storage market to capture the commensurate grid benefits. A new V2G-specific program offering, with program terms that are more suitable for this technology and, potentially, based on CMCP implementation, is critically needed to deploy V2G projects in New York. Additionally, a V2G-specific program would allow the Commission to modify program terms and incentive levels more easily as lessons are learned and the industry matures without having to modify the underlying VDER tariff, which applies to all distributed energy resources.

Conclusion

VGIC appreciates the opportunity to provide these comments and looks forward to working with the utilities, the Commission, and other stakeholders to ensure the success of New York's transportation electrification efforts.

Respectfully submitted,

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